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Systems and methods for determining an articulated trailer angle

Abstract

Systems and methods determine an angle of an articulated trailer relative to a tractor that the trailer is hitched to. An encoder is positioned beneath a fifth-wheel of a tractor to couples with a kingpin of the trailer when the trailer is hitched to the tractor. The coupler has a rotating shaft that may include pins that physically interact with the kingpin and/or may include a magnet that magnetically attaches to the kingpin. A clearance and cleaning block may be positioned on the spring plate to interact with a bottom surface of a kingpin of the trailer during hitching of the trailer to the tractor. A LIDAR attached to the tractor may detect a front end of the trailer to determine the trailer angle relative to the tractor.

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Background/Summary

RELATED APPLICATIONS (1) This patent application is a continuation of U.S. patent application Ser. No. 17/848,188, filed Jun. 23, 2022, which claims priority to U.S. Provisional Patent Application No. 63/214,227, filed on Jun. 23, 2021, and claims priority to U.S. Provisional Patent Application No. 63/327,723, filed on Apr. 5, 2022. The disclosure of each of the prior applications is incorporated herein by reference in its entirety.

BACKGROUND

(1) Trucks are an essential part of modern commerce. These trucks transport materials and finished goods across the continent within their large interior spaces. Such goods are loaded and unloaded at various facilities that can include manufacturers, ports, distributors, retailers, and end users. The start and end locations are referred to as “yards” and include areas that trailers are parked (and/or staged) and moved to and from for access by tractors (trucks) for loading to a dock door for loading/unloading cargo into the associated facility, leaving the yard for travel to its destination, or entering the yard from its destination. Autonomous yard vehicles technology includes tractors (trucks) that are capable of automatically (without human intervention, or with human intervention via teleoperation) coupling, decoupling, and maneuvering trailers that are within the yard.

(2) Safety is of utmost importance in such automated yards. The automatic maneuvering of said trailers results in situations where, if a person or other obstacle is in the intended path of the trailer or tractor, because there is no human operating the tractor, there are situations where the tractor may not know of a human or obstacle. Thus, additional sensors are desired so that the controller of the automated tractor can maneuver the trailers safely.

(3) Additional difficulties arise because various manufactures and freight companies have their own trailers. Thus, while an automated yard vehicle may have associated sensors, it is difficult to utilize sensors on the trailers themselves because it requires human (or machine) intervention on the trailer prior to maneuvering the trailer. This additional intervention step is timely and creates an additional location for safety concern.

SUMMARY

(4) Trucks are an essential part of modern commerce. These trucks transport materials and finished goods across the continent within their large interior spaces. Such goods are loaded and unloaded at various facilities that can include manufacturers, ports, distributors, retailers, and end users. Large over-the road (OTR) trucks typically consist of a tractor or cab unit and a separate detachable trailer that is interconnected removably to the cab via a hitching system that consists of a so-called fifth wheel and a kingpin.

(5) Further challenges in trucking relate to docking, loading and unloading of goods to and from trailers. Warehouses and good distribution facilities have yards with multiple loading docks, and the trailer is positioned at one of the loading docks for loading and unloading. In an automated yard, the OTR truck stops at a designated location in staging area of the yard, and the OTR tractor

detaches, leaving the trailer at the designated location. An autonomous tractor moves the trailer to a first one of the loading docks for unloading and/or loading. Another, or the same, autonomous tractor moves the trailer away from the loading dock when loading and/or unloading is complete and parks the trailer in a designated location of the staging area. The trailer may also be moved between loading docks if needed by another, or the same, autonomous tractor. Another, or the same, OTR tractor couples with the trailer and the OTR truck departs the yard for another destination.

(6) One aspect of the present embodiments includes the realization that for an autonomous tractor to reverse an articulated trailer accurately and safely into a designated location, such as a loading dock, the autonomous tractor requires accurate knowledge of a position and/or location and/or orientation of the back end of the articulated trailer always. However, the articulated trailer does not have sensors for determining this information. The present embodiments solve this problem by determining an angle between the articulated trailer and the autonomous tractor, and then extrapolating a location of the back end of the articulated trailer based on a location of the autonomous tractor, an orientation of the autonomous tractor, a length of the articulated trailer and the angle between the articulated trailer and the autonomous tractor.

(7) In certain embodiments, a trailer angle encoder for determining an angle between a tractor and a trailer coupled thereto includes an arm coupled at a pivot with a flange located beneath a fifth-wheel of the tractor and an optical encoder positioned at a first end of the arm and having a rotatable shaft with a mechanical coupler. The arm being positioned to mechanically couple the mechanical coupler with a kingpin of the trailer.

(8) In certain embodiments, a method for determining an angle between a tractor and a trailer that are coupled together includes: controlling, from a controller of the tractor, the tractor to pull the trailer a short distance; determining, from an optical encoder mounted on the tractor and mechanically coupled with the trailer, a change in angle between the tractor and the trailer; and calculating the angle between the tractor and the trailer based on the change in angle.

(9) In certain embodiments, a trailer angle encoder for determining an angle between a tractor and a trailer coupled thereto includes a spring plate for coupling at a first end with an underside of a fifth-wheel of the tractor, an optical encoder attached to the spring plate, a magnet mounted to a rotatable shaft of the optical encoder, and a clearance and cleaning block positioned on the spring plate to interact with a bottom surface of a kingpin of the trailer during hitching of the tractor to the trailer; wherein the magnet magnetically couples with the bottom surface of the kingpin when the tractor is hitched to the trailer.

(10) In certain embodiments, a software product includes instructions, stored on non-transitory computer-readable media, wherein the instructions, when executed by a processor, perform steps for determining an angle between a tractor and a trailer that are coupled together, the software product including instructions for controlling, from a controller of the tractor, the tractor to pull the trailer a short distance; instructions for determining, from an optical encoder mounted on the tractor and mechanically coupled with the trailer, a change in angle between the tractor and the trailer; and instructions for calculating the angle between the tractor and the trailer based on the change in angle.

(11) In certain embodiments, a method for determining an angle between a tractor and a trailer that are coupled together includes: capturing, within a controller of the tractor, a point cloud using a rear facing LIDAR positioned on the tractor; converting points of the point cloud corresponding to front corners of the trailer to coordinate form; and calculating the angle between the tractor and the trailer based upon the coordinates of the front corners of the trailer.

(12) In certain embodiments, a software product includes instructions, stored on non-transitory computer-readable media, wherein the instructions, when executed by a processor, perform steps for determining an angle between a tractor and a trailer that are coupled together, the software product including instructions for capturing, within a controller of the tractor, a point cloud using a rear facing LIDAR positioned on the tractor; instructions for converting points of the point cloud

corresponding to front corners of the trailer to coordinate form; and instructions for calculating the angle between the tractor and the trailer based upon the coordinates of the front corners of the trailer.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is an aerial view showing one example autonomous yard that uses an autonomous tractor to move trailers between a staging area and loading docks of a warehouse, in embodiments.
- (2) FIG. 2 is a block diagram illustrating key functional components of the autonomous tractor of FIG. 1, in embodiments.
- (3) FIG. 3 is a side elevation showing the tractor of FIG. 1 reversing under a lower surface of a trailer, in embodiments.
- (4) FIG. 4 shows one example hitch and unhitch sequence of states implemented by the function state machine of FIG. 2 for coupling and uncoupling the tractor and the trailer, in embodiments.
- (5) FIG. 5 shows the trailer angle module of FIG. 2 in further example detail, in embodiments.
- (6) FIG. 6 is a schematic plan view illustrating the tractor of FIGS. 1-3 backing the trailer up to one of a plurality of adjacent loading docks of the warehouse of FIG. 1, in embodiments.
- (7) FIG. 7 is a schematic showing example assumptions made by the motion planner running in the maneuvering module of the tractor to determine the steering angle for controlling the tractor to reverse the trailer, in embodiments.
- (8) FIG. 8 is a schematic diagram showing one example trailer angle encoder positioned beneath the FW of the tractor of FIGS. 1-3 and in a disengaged position as the tractor couples with the trailer, in embodiments.
- (9) FIG. 9 is a schematic showing the trailer angle encoder of FIG. 8 in an engaged position to mechanically couple with the kingpin of the trailer while the tractor and trailer are coupled together, in embodiments.
- (10) FIGS. 10-12 are schematic diagrams showing another example trailer angle encoder positioned beneath the FW of the tractor of FIGS. 1-3, in embodiments.
- (11) FIG. 13 shows the LIDAR of FIG. 2 mounted on a cab portion of the tractor to face the trailer, in embodiments.
- (12) FIG. 14 is a schematic illustrating example operation of the LIDAR of FIG. 13 to detect front corners of the trailer when the trailer is in-line with the tractor, in embodiments.
- (13) FIG. 15 is a schematic illustrating example operation of the LIDAR of FIG. 13 to detect the front corners of the trailer when the trailer is at a twenty-degree angle to the tractor, in embodiments.
- (14) FIG. 16 is a front perspective view of the tractor coupled with the trailer of FIG. 1, showing a LIDAR mounted to an extensible mast, in embodiments.
- (15) FIG. 17 is a schematic diagram showing one example magnetically coupled trailer angle encoder positioned beneath the FW of the tractor prior to the tractor coupling (hitching) with the trailer, in embodiments.
- (16) FIG. 18 is a schematic showing the magnetically coupled trailer angle encoder of FIG. 17 magnetically coupled with the kingpin of the trailer hitched to the tractor, in embodiments.
- (17) FIGS. 19, 20 and 21 shows side, top and front elevations, respectively, of the spring plate of FIGS. 17 and 18, in embodiments.
- (18) FIG. 22 is a top view of the magnetically coupled trailer angle encoder of FIGS. 17 and 18, in embodiments.
- (19) FIGS. 23, 24, and 25 are front, side, and top elevations, respectively, of the two-axis positioning table of FIGS. 17 and 18, in embodiments.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(20) In an automated yard, an autonomous tractor moves trailers between staging areas and loading docks for unloading and/or loading. The autonomous tractor repeatedly couples (hitches) to a trailer, moves the trailer, and then decouples (unhitches) from the trailer.

(21) FIG. 1 is an aerial view showing one example autonomous yard **100** (e.g., a goods handling facility, shipping facility, etc.) that uses an autonomous tractor **104** to move trailers **106** between a staging area **130** and loading docks of a warehouse **110**. For example, an over-the-road (OTR) tractors **108** deliver goods-laden trailers **106** from remote locations and retrieve trailers **106** for return to such locations (or elsewhere-such as a storage depot). In a standard operational procedure, OTR tractor **108** arrives with trailer **106** and checks-in at a facility entrance checkpoint **109**. A guard/attendant enters information (e.g., trailer number or QR (ID) code scan-embedded information already in the system, which would typically include: trailer make/model/year/service connection location, etc.) into a mission controller **102** (e.g., a computer software server that may be located offsite, in the cloud, fully onsite, or partially located within a facility building complex, shown as a warehouse **110**). Warehouse **110** includes perimeter loading docks (located on one or more sides of the building), associated (typically elevated) cargo portals and doors, and floor storage, all arranged in a manner familiar to those of skill in shipping, logistics, and the like.

(22) By way of a simplified operational example, after arrival of OTR tractor **108** and trailer **106**, the guard/attendant at checkpoint **109** directs the driver to deliver trailer **106** to a specific numbered parking space in a designated staging area **130**, which may include a large array of side-by-side trailer parking locations, arranged as appropriate for the facility's overall layout.

(23) Once the driver has parked the trailer in the designated parking space of the staging area **130**, he/she disconnects the service lines and ensures that connectors are in an accessible position (i.e. if adjustable/sealable), and decouples OTR tractor **108** from trailer **106**. If trailer **106** is equipped with swing doors, this can also provide an opportunity for the driver to unlatch and clip trailer doors in the open position, if directed by yard personnel to do so.

(24) At some later time, (e.g., when warehouse is ready to process the loaded trailer) mission controller **102** directs (e.g., commands or otherwise controls) tractor **104** to automatically couple (e.g., hitch) with trailer **106** at a pick-up spot in staging area **130** and move trailer **106** to a drop-off spot at an assigned unloading dock in unloading area **140** for example. Accordingly, tractor **104** couples with trailer **106** at the pick-up spot, moves trailer **106** to unloading area **140**, and then backs trailer **106** into the assigned loading dock at the drop-off spot such that the rear of trailer **106** is positioned in close proximity with the portal and cargo doors of warehouse **110**. The pick-up spot and the drop-off spot may be any designated trailer parking location in staging area **130**, any loading dock in unloading area **140**, and any loading dock within loading area **150**.

(25) Manual and/or automated techniques are used to offload the cargo from trailer **106** and into warehouse **110**. During unloading, tractor **104** may remain hitched to trailer **106** or may decouple (e.g., unhitch) to perform other tasks. After unloading, mission controller **102** directs tractor **104** to move trailer **106** from a pick-up spot in unloading area **140** and to a drop-off spot, either returning trailer **106** to staging area **130** or delivering trailer **106** to an assigned loading dock in a loading area **150** of warehouse **110**, where trailer **106** is then loaded. Once loaded, mission controller **102** directs tractor **104** to move trailer **106** from a pick-up spot in loading area **150** to a drop-off spot in staging area **130** where it may await collection by another (or the same) OTR tractor **108**. Given the pick-up spot and the drop-off spot, tractor **104** may autonomously move trailer **106**.

(26) FIG. 2 is a block diagram illustrating key functional components of tractor **104**. Tractor **104** includes a battery **202** for powering components of tractor **104** and a controller **206** with at least one digital processor **208** communicatively coupled with memory **210** that may include one or both of volatile memory (e.g., RAM, SRAM, etc.) and non-volatile memory (e.g., PROM, FLASH, Magnetic, Optical, etc.). Memory **210** stores a plurality of software modules including machine-readable instructions that, when executed by the at least one processor **208**, cause the at least one

processor **208** to implement functionality of tractor **104** as described herein to operate autonomously within autonomous yard **100** under direction from mission controller **102**.

(27) When tractor **104** is an electric tractor, tractor **104** also includes at least one drive motor **212** controlled by a drive circuit **214** to mechanically drive a plurality of wheels (not shown) to maneuver tractor **104**. Drive circuit **214** includes a safety feature **215** that deactivates motion of tractor **104** when it detects that rotation of drive motor **212** is impeded (e.g., stalled) and that drive motor **212** is drawing a current at or greater than a stalled threshold (e.g., above one of **400A**, **500A**, **600A**, **700A**, etc. depending on the configuration of the drive motor **212**), for a predetermined period (e.g., five seconds). Safety feature **215** may thereby prevent damage to tractor **104** and/or other objects around tractor **104** when tractor **104** is impeded by an object. Safety feature **215** is described above with respect to an electric tractor. It should be appreciated that a similar safety feature could be included for diesel-based tractors, such as reducing engine power when an RPM threshold goes above a pre-set threshold. When safety feature **215** is tripped, tractor **104** requires manual reactivation before being able to resume movement. Accordingly, tripping safety feature **215** is undesirable.

(28) Tractor **104** also includes a location unit **216** (e.g., a GPS receiver) that determines an absolute location and orientation of tractor **104**, a plurality of cameras **218** for capturing images of objects around tractor **104**, and at least one Light Detection and Ranging (LIDAR) device **220** (hereinafter LIDAR **220**) for determining a point cloud about tractor **104**. Location unit **216**, the plurality of cameras **218**, and the at least one LIDAR **220** cooperate with controller **206** to enable autonomous maneuverability and safety of tractor **104**. Tractor **104** includes a fifth wheel (FW) **222** for coupling with trailer **106** and a FW actuator **224** controlled by controller **206** to position FW **222** at a desired height. In certain embodiments, FW actuator **224** includes an electric motor coupled with a hydraulic pump that drives a hydraulic piston that moves FW **222**. However, FW actuator **224** may include other devices for positioning FW **222** without departing from the scope hereof. Tractor **104** may also include an air actuator **238** that controls air supplied to trailer **106** and a brake actuator **239** that controls brakes of tractor **104** and trailer **106** when connected thereto via air actuator **238**.

(29) Controller **206** also includes a trailer angle module **232** that determines a trailer angle **233** between tractor **104** and trailer **106** based on one or both of a trailer angle measured by an optical encoder **204** positioned near FW **222** and mechanically coupled with trailer **106** and a point cloud **221** captured by the at least one LIDAR **220**.

(30) Controller **206** may implement a function state machine **226** that controls operation of tractor **104** based upon commands (requests) received from mission controller **102**. For example, mission controller **102** may receive a request (e.g., via an API, and/or via a GUI used by a dispatch operator) to move trailer **106** from a first location (e.g., slot X in staging area **130**) to a second location (e.g., loading dock Y in unloading area **140**). Once this request is validated, mission controller **102** invokes a mission planner (e.g., a software package, not shown) that computes a 'mission plan' for each tractor **104**. For example, the mission plan is an ordered sequence of high level primitives to be followed by tractor **104**, in order to move trailer **106** from location X to location Y. The mission plan may include primitives such as drive along a first route, couple with trailer **106** in parking location X, drive along a second route, back trailer **106** into a loading dock, and decouple from trailer **106**.

(31) Function state machine **226** includes a plurality of states, each associated with at least one software routine (e.g., machine-readable instructions) that is executed by processor **208** to implements a particular function of tractor **104**. Function state machine **226** may transitions through one or more states when following the primitives from mission controller **102** to complete the mission plan.

(32) Controller **206** may also include an articulated maneuvering module **240**, implemented as machine-readable instructions that, when executed by processor **208**, cause processor **208** to controls drive circuit **214** and steering actuator **225** to maneuver tractor **104** based on directives

from mission controller **102**.

(33) Controller **206** may also include a navigation module **234** that uses location unit **216** to determine a current location and orientation of tractor **104**. Navigation module **234** may also use other sensors (e.g., camera **218** and/or LIDAR **220**) to determine the current location and orientation of tractor **104** using dead-reckoning techniques.

(34) FIG. **3** is a side elevation showing tractor **104** of FIG. **1** reversing under a lower surface **302** of trailer **106**. FIG. **4** shows one example hitch sequence **400** of states implemented by function state machine **226** of tractor **104**, FIGS. **1-3**, for coupling tractor **104** with trailer **106**, and one example unhitch sequence **450** of states implemented by function state machine **226** for decoupling tractor **104** from trailer **106**. FIG. **4** also shows example transitions between sequences when alignment fail is detected (e.g., when an activity of the current state fails for some reason), which allows function state machine **226** to recover from the failure (e.g., undo certain actions) and to reattempt the command. FIGS. **3** and **4** are best viewed together with the following description.

(35) As shown in FIG. **3**, landing gear **306** of trailer **106** is sufficiently extended such that a lower surface **302** (e.g., a FW plate) of a front end of trailer **106** is high enough above ground level to allow FW **222**, when fully retracted, to be pushed thereunder without stalling drive motor **212** of tractor **104**. That is, drive motor **212** provides sufficient force to push FW **222** under lower surface **302**. However, landing gear **306** is extended by a driver of OTR tractor **108** when leaving trailer **106** in staging area **130** of autonomous yard **100**, and therefore the height of lower surface **302** is at the discretion of the driver and may not be consistent between trailers **106**. Further, the force required to move FW **222** under lower surface **302** is also dependent upon a weight (e.g., of goods) at the front end of trailer **106**. When drive motor **212** is unable to provide sufficient force to push FW **222** beneath lower surface **302**, such as when landing gear **306** is not sufficiently extended, drive motor **212** stalls.

(36) In response to receiving a hitch command from mission controller **102**, once tractor **104** is aligned with trailer **106**, controller **206**, in state **402**, stows FW **222** and controls drive circuit **214** to move tractor **104** slowly backwards as indicated by arrow **304**. When controller **206** detects that FW **222** is beneath lower surface **302** of trailer **106**, drive motor **212** is stopped and function state machine **226** transitions to state **404**. If controller **206** determines that tractor **104** is not correctly aligned with trailer **106**, function state machine **226** transitions to state **458** of unhitch sequence **450** such that another attempt may be made. In state **404**, controller **206** controls FW actuator **224** to lift trailer **106** and controls drive circuit **214** to back tractor **104**, and thus FW **222**, up to a kingpin **308** of trailer **106**. In state **406**, controller **206** controls FW actuator **224** to raise FW **222** and thereby lift the front end of trailer **106** for Trailer Connect (e.g., a process of connecting air lines/electrical from tractor **104** to trailer **106** using gladhand ID and orientation). In state **408**, controller **206** controls drive circuit **214** to perform a tug test. If controller **206** determines that tractor **104** is not correctly coupled with trailer **106** (e.g., the kingpin did not latch), function state machine **226** transitions to state **458** of unhitch sequence **450** such that another attempt may be made. In state **410**, controller **206** controls trailer air actuator **238** to perform the TC connect. If controller **206** determines that the TC did not connect successfully, function state machine **226** transitions to state **454** of unhitch sequence **450** such that another attempt may be made. In state **412**, controller **206** controls trailer air actuator **238** to supply trailer air and controls FW actuator **224** to raise FW **222** higher to ensure that the trailer landing gear clears the ground in preparation to drive.

(37) In response to receiving an unhitch command from mission controller **102**, once trailer **106** is correctly positioned, controller **206**, in state **452**, controls trailer air actuator **238** to release trailer air and controls FW actuator **224** to lower FW **222** and the front end of trailer **106**. In state **454**, controller **206** controls trailer air actuator **238** to disconnect the TC from trailer **106**. In state **456**, controller **206** controls drive circuit **214** to move tractor **104** forward to perform a tug test. In state **458**, controller **206** controls FW actuator **224** to lower the front end of trailer **106** to the ground. In

state **460**, controller **206** controls FW actuator **224** to unlatch from the trailer kingpin. In state **462**, controller **206** controls FW actuator **224** to stow FW **222** and controls drive circuit **214** to cause tractor **104** to move forward away from trailer **106**.

(38) Trailer Angle Measurement

(39) FIG. 5 shows trailer angle module **232** of controller **206**, FIG. 2, in further example detail. Trailer angle module **232** includes a LIDAR angle estimator **510** for determining a LIDAR angle **512** that estimates an angle of trailer **106** relative to tractor **104** based upon point cloud **221** captured by LIDAR **220** and including points corresponding to a front end of trailer **106**. Trailer angle module **232** also includes an initial angle estimator **520** that includes algorithms for determining an optical trailer angle **522** based on angular change **502** output by trailer angle encoder **204** over a short movement of tractor **104** and trailer **106** (e.g., shortly after coupling/hitching of trailer **106** to tractor **104**). Trailer angle module **232** also includes an encoder estimator **530** that updates current trailer angle **233** based on angular change **502**. Encoder estimator **530** may include a Kalman filter **532** that processes LIDAR angle **512** and angular change **502** to generate current trailer angle **233**. For example, Kalman filter **532** reduces noise on LIDAR angle **512**. Kalman filter **532** estimates the unknown initial bias in optical trailer angle **522** (e.g., in incremental encoder data from angle encoder **204**) based upon LIDAR angle **512** determined from LIDAR **220** (described below). The estimated bias is refined and removed from angular change **502** resulting in a low noise optical trailer angle **522**. Further biases due to crooked tandems and/or trailers may be removed using short move calculations, described below.

(40) FIG. 6 is a schematic plan view illustrating tractor **104** backing trailer **106** up to one loading dock **602(2)** of a plurality of adjacent loading docks **602(1)-(5)** of warehouse **110**. Tractor **104** coupled with trailer **106** may be referred to as vehicle **612**. Each loading dock **602(1)-(5)** has a corresponding loading door **604(1)-(5)**, and as shown, one trailer **106(2)** is parked at loading dock **602(3)**. Since trailer doors are at the rear of trailer **106**, trailer **106** is reversed up to loading dock **602** and is correctly aligned with loading door **604** to provide full and safe access to trailer **106**. A reference path **606** may be defined for loading dock **602(2)** to facilitate alignment of trailer **106(1)** into loading dock **602(2)**. Maneuvering module **240** may predict a path **608** of trailer **106(1)** when determining a steering angle **250** for steering wheels **610** of tractor **104**.

(41) FIG. 7 is a schematic showing example assumptions made by maneuvering module **240** of tractor **104** when determining steering angle **250** for controlling tractor **104** to reverse trailer **106**. For purposes of simplification, tractor **104** and trailer **106** are approximated in a kinematic bicycle model with nonholonomic constraints. A front axle **702** of tractor **104** is approximated by a single steerable modelled wheel **704** at the axle's center, a rear axle **706** of tractor **104** is approximated by a single non-steering modelled wheel **708** at the axle's center, and tandem **710** of trailer **106** is approximated as a single non-steering modelled wheel **712** centered in between both axles of tandem **710**. This simplified representation of tractor **104** and trailer **106** allows any complex dynamic interactions between the actual wheels to be ignored and the nonholonomic constraint implies that none of the actual wheels move laterally. As shown in FIG. 7, tractor **104** is assumed to move along a circle **714** perpendicular to non-steering modelled wheel **708** about a tractor center of rotation **716**, and trailer **106** is assumed to travel along a circle **718** perpendicular to non-steering modelled wheel **712** about a trailer center of rotation **720**. This assumption is generally safe when tractor **104** and trailer **106** are moving at low speed (e.g., less than 15 miles-per-hour). Further assumptions include: tires do not deform, tires along an axle are properly aligned, and motion of the steering wheels of front axle **702** is approximated by the average angle the wheels.

(42) However, to accurately back trailer **106** into loading dock **602(2)**, tractor **104** requires accurate knowledge of the position of the back end of trailer **106**, and non-steering modelled wheel **712**, relative to tractor **104**.

(43) FIG. 8 is a schematic diagram showing one example trailer angle encoder **800** positioned beneath FW **222** of tractor **104** and in a disengaged position as tractor **104** couples (hitches) with

trailer **106**. Trailer angle encoder **800** may represent trailer angle encoder **204** of FIG. 2. FIG. 9 is a schematic showing trailer angle encoder **800** of FIG. 8 in an engaged position to mechanically couple with kingpin **308** of trailer **106** while tractor **104** and trailer **106** are coupled together. Trailer angle encoder **800** measures trailer angle by mechanically coupling with kingpin **308** of trailer **106**. Trailer angle encoder **800** includes an arm **802** attached via a pivot **804** to an existing flange **806** of FW **222**. An optical encoder **808** is positioned at a first end, away from pivot **804**, of arm **802** such that it is positioned beneath, and pivoted away from, locking jaws **810** of FW **222**. Trailer angle encoder **800** includes a disc **812** mechanically coupled with an input spindle of optical encoder **808** and having a plurality (e.g., three) vertical pins **814** (e.g., spikes, teeth, knife blades perpendicular to a lower surface of kingpin **308**, etc.) positioned around an upper surface of disc **812**. In the example of FIG. 8, a solenoid **816** operates to compress a spring **818** and pull a second end, opposite the first end, of arm **802** such that disc **812** and pins **814** are retracted away from locking jaws **810** as tractor **104** moves beneath trailer **106**, as indicated by arrow **820**. Accordingly, activating solenoid **816** moves disc **812**, pins **814**, and optical encoder **808** away from FW **222** as kingpin **308** is captured by locking jaws **810**, thereby preventing damage to trailer angle encoder **800** during coupling and decoupling of trailer **106** from tractor **104**. In other embodiments, operation of solenoid **816** and spring **818** may be reversed, whereby solenoid **816** is activated to move disc **812**, pins **814**, and optical encoder **808** towards FW **222** when tractor **104** and trailer **106** are coupled together. Other actuators may be used in place of solenoid **816** and spring **818** to move arm **802** without departing from the scope hereof.

(44) FIG. 9 shows trailer angle encoder **800** of FIG. 8 mechanically coupled with kingpin **308** captured by locking jaws **810** of FW **222**. With kingpin **308** captured by locking jaws **810**, solenoid **816** is deactivated, allowing spring **818** to push down on the second end of arm **802**, which pivots about pivot **804** such that the first end of arm **802** presses pins **814** into a bottom surface (e.g., of the kingpin flange) of kingpin **308**. Pins **814** may be individually spring loaded to ensure good contact when the lower surface of kingpin **308** is uneven, and are durable (e.g., made of a hardened steel or titanium) and mechanically couple with the lower surface of kingpin **308** to cause disc **812** and input shaft of optical encoder **808** to rotate as kingpin **308** rotates relative to tractor **104**. For example, the input shaft of optical encoder **808** has minimal rotational resistance, and thereby follows movement of kingpin **308** through contact of pins **814**. Advantageously, once trailer **106** is coupled with of tractor **104**, solenoid **816** is deactivated and spring **818** maintains pressure against arm **802** and pins **814** maintain contact with kingpin **308**.

(45) FIGS. 10-12 are schematic diagrams showing another example trailer angle encoder **1000** positioned beneath FW **222** of tractor **104** of FIG. 1. Trailer angle encoder **1000** may represent trailer angle encoder **204** of FIG. 2. FIG. 10 shows trailer angle encoder **1000** in a disengaged position as tractor **104** couples (hitches) with trailer **106** and FIG. 11 shows trailer angle encoder **1000** of FIG. 10 in an engaged position to mechanically coupled with kingpin **308** of trailer **106** while tractor **104** and trailer **106** are coupled together. FIG. 12 shows a longitudinally and laterally adjustable arm **1002** of trailer angle encoder **1000** in further detail. Trailer angle encoder **1000** is similar to trailer angle encoder **800** of FIGS. 8 and 9, except that trailer angle encoder **1000** includes a conical adapter **1012**, in place of disc **812** and pins **814**, for mechanically coupling with kingpin **308**, and arm **802** is replaced by longitudinally and laterally adjustable arm **1002**. Accordingly, only difference between trailer angle encoder **1000** and trailer angle encoder **800** will be described.

(46) Conical adapter **1012** couples with the input shaft of optical encoder **808** and has a conical shape that tapers internally from a first diameter, nearest optical encoder **808**, smaller than the diameter of the flange of kingpin **308** to a second diameter greater than the diameter of the flange of kingpin **308**. Conical adapter **1012** is at least partially formed of a flexible material (e.g., rubber, polyurethane, oil resistant room-temperature-vulcanizing (RTV) silicone, etc.) that mechanically grips the flange of kingpin **308** when pressed there against to cause the input shaft of optical

encoder **808** to rotate as kingpin **308** rotates relative to tractor **104**. As shown in FIG. **10**, solenoid **816** is activated to move conical adapter **1012** out of the path of kingpin **308** as tractor **104** couples with trailer **106**. As shown in FIG. **11**, once tractor **104** and trailer **106** are coupled, solenoid **816** is deactivated and conical adapter **1012** is pressed against the flange of kingpin **308** by spring **818**.

(47) As shown in FIG. **12**, longitudinally and laterally adjustable arm **1002** allows both longitudinal and lateral movement to allow conical adapter **1012** to center on kingpin **308**. Arm **1002** is formed of a first plate **1202** that forms an aperture **1204** for securing optical encoder **808** and a second plate **1206** that pivotably couples with pivot **804**. Plates **1202** and **1206** are joined by two longitudinal shafts **1208(1)** and **1208(2)** that allow longitudinal movement, indicated by arrow **1205**, of plate **1202** relative to plate **1206**. Two springs **1210(1)**, **1210(2)** are positioned on longitudinal shaft **1208(1)**, one on each side of at least part of plate **1202**. Similarly, another two springs **1210(3)**, **1210(4)** are positioned on longitudinal shaft **1208(2)**, one on each side of at least part of plate **1202**. Accordingly, springs **1210** cause plate **1202** to return to a nominal longitudinal center position when conical adapter **1012** is decoupled from kingpin **308**. Pivot **804** allows both rotation of plate **1206** around pivot **804** and lateral movement of plate **1206** along pivot **804**. Two springs **1212(1)** and **1212(2)** are positioned on pivot **804**, one on each side of at least part of plate **1206** and cause second plate **1206** to return to a nominal lateral center position when conical adapter **1012** is decoupled from kingpin **308**.

(48) In one embodiment, optical encoder **808** is a quadrature optical encoder that generates angular change **205** (e.g., a count of pulses that indicate the changing angle over time) as kingpin **308** (e.g., trailer **106**) rotates relative to tractor **104**. For example, with θ being an angle of trailer **106** relative to tractor **104**, the quadrature encoder counter output with a 1024-line encoder is given by:

$$(49) \quad \theta = \theta_{sub.0} + \frac{360}{4096} \text{enc_count}$$

where $\theta_{sub.0}$ is estimated by a Kalman filter.

(50) The standard deviation of the quadrature encoder's quantization noise is:

$$(51) \quad \frac{1}{\sqrt{12}} = \frac{\frac{360}{4096}}{\sqrt{12}} = 0.02537^\circ \approx \frac{1}{40} \text{degrees}$$

and the signal to noise quantization ration is:

$$(52) \quad 20 \log_{10}(2^{14}) = 84 \text{dB}$$

(53) Accordingly, optical encoder **808** provides a high-resolution measurement of change in the angle between trailer **106** and tractor **104**. However, since the angle between tractor **104** and trailer **106** is unknown when tractor **104** couples with trailer **106**, the relative change in angular position provided in angular change **502** by optical encoder **808** cannot indicate an absolute angle between tractor **104** and trailer **106**.

(54) Determining Trailer Angle Using LIDAR

(55) FIG. **13** shows LIDAR **220** mounted on a cab portion **1300** of tractor **104** to face trailer **106**. Kingpin **308** of trailer **106** is captured by FW **222**, and is therefore a point of rotation of trailer **106** relative to tractor **104**. FIG. **14** is a schematic illustrating example operation of LIDAR **220** to detect front corners **1302** and **1304** of trailer **106** when trailer **106** is in-line with tractor **104**. FIG. **15** is a schematic illustrating example operation of LIDAR **220** to detect front corners **1302** and **1304** of trailer **106** when trailer **106** is at a twenty-degree angle to tractor **104**. As shown, corners **1302** and **1304** (e.g., the front of trailer **106**) rotate about kingpin **308**. FIGS. **13**, **14**, and **15** are best viewed together with the following description.

(56) LIDAR **220** generates point cloud **221** by scanning at least one beam **1402** of light and detecting light reflected from objects in the path of the light. LIDAR **220** is positioned on a rear facing surface of a cab portion **1300** of tractor **104** and thereby scans beam **1402** across at least a front end of trailer **106**. Each point in point cloud **221** defines a distance r and an angle θ . The following discussion uses n -row column vectors ($n \geq 2$), and M and b scalars. When beam **1402** misses the front end of trailer **106**, r has a high value, and thus the data may be ignored for purposes of trailer angle calculations. Accordingly, beams **1402** corresponding to corners **1302** and

1304 of trailer **106** may be easily identified in data from LIDAR **220**.

(57) The $[r, \theta]$ LIDAR data (e.g., point cloud **221**) is converted into $[x, Y]$ coordinate form, such that:

$$(58) Y = M \cdot \text{Math. } x + b, Y = [x[1]] \begin{bmatrix} M \\ b \end{bmatrix}, \text{ and } \begin{bmatrix} M \\ b \end{bmatrix} = ([x[1]]^T [x[1]])^{-1} [x[1]]^T Y$$

Giving:

$$(59) \text{TrailerAngle} = \tan^{-1}(M)$$

(60) LIDAR angle estimator **510** includes at least one algorithm that implements at least part of the equations described above to determine LIDAR angle **512** based upon point cloud **221**.

Advantageously, LIDAR angle estimator **510** generates LIDAR angle **512** independently of specific dynamics (e.g., length) trailer **106**. In certain embodiments, LIDAR angle estimator **510** may be invoked at intervals and/or on demand to generate LIDAR angle **512** to check current trailer angle **233** generated by other means, and/or may provide input to update of current trailer angle **233**.

(61) Determining Trailer Angle Through Short Movement

(62) In an alternative embodiment, an initial angle of trailer **106** is estimated using a Kinematic equation model, a short move of tractor **104** and trailer **106**, and measured change in trailer angle by trailer angle encoder **204**. The Kinematic equation model is defined as follows. Circular motion is defined as:

$$(63) v = r \cdot \dot{\theta}$$

giving

$$(64) \dot{\theta} = \frac{v}{r}$$

where r is the circle's radius

$$(65) (\text{radius of curvature}, \text{curvature} = \frac{1}{r}).$$

Kinematic equations of tractor **104** and trailer **106** are defined as:

$$(66) \dot{x} = v(t) \cos(\theta(t)) \dot{y} = v(t) \sin(\theta(t)) \dot{\phi} = \frac{v(t)}{r} = \frac{v(t)}{L} \tan(\alpha(t)) \dot{\theta} = \frac{v(t)}{r} = -\frac{v(t)}{L_2} \sin(\theta(t) - \phi(t))$$

where x, y is position, ϕ is a heading of tractor **104**, and θ is a heading of trailer **106**.

(67) Solving for steering angle:

$$(68) \theta(t) = \tan^{-1}\left(\frac{L}{r}\right)$$

and solving for trailer angle (e.g., angle between tractor **104** and trailer **106**):

$$(69) \theta(t) - \phi(t) = \sin^{-1}\left(-\frac{L_2}{r}\right)$$

where L is the length of tractor **104** (e.g., the distance between single steerable modelled wheel **704** and single non-steering modelled wheel **708**), and $L_{\text{sub.2}}$ is the length of trailer **106** (e.g., the distance between kingpin **308** and single non-steering modelled wheel **712**).

(70) Accordingly, the radius of curvature of trailer **106** is:

$$(71) r_{\text{trailer}} = \sqrt{r^2 - L_2^2}$$

(72) Initial angle estimator **520**, FIG. 5, may implement at least one algorithm based on the above equations to determine optical trailer angle **522** based on angular change **502**. Advantageously, initial angle estimator **520** operates independently of trailer specifics such as length of trailer **106**. In one example of operation, initial angle estimator **520** reads trailer angle encoder **204** (e.g., to reset the quadrature encoder counter), controls tractor **104** to move a short distance, and then reads an angular change **502** from optical encoder **808**. To ensure any angular change is not lost in quantization noise, the short distance moved should be at least one meter. For example, an angular change of two-and-a-half degrees would require a one-meter movement for a twenty-five-meter radius turn. Initial angle estimator **520** then calculates a current angle of trailer **106** relative to tractor **104** based on the distance moved ΔS , angular change **502**, and the following equations.

$$(73) \theta_0 = -\frac{\Delta S}{L_2} \sin(\theta_0 + \theta - \theta_0 - \theta_0)$$

giving

$$(74) \theta_0 = \theta + \theta_0 - \theta_0 + \sin^{-1}\left(-\frac{L_2}{\Delta S}\right)$$

(75) Initial angle estimator **520** may be invoked after tractor **104** couples with trailer **106** to determine optical trailer angle **522** that is used to determine an initial value for current trailer angle **233**. Thereafter, encoder estimator **530** updates current trailer angle **233** based on angular changes **502** from trailer angle encoder **204** (e.g., from optical encoder **808**). In certain embodiments, initial angle estimator **520** is invoked at intervals, or on demand, to validate and/or update current trailer angle **233**.

(76) In certain embodiments, encoder estimator **530** may process both angular change **502** and LIDAR angle **512** to update current trailer angle **233**. Encoder estimator **530** may also process input from other means of determining current trailer angle **233**, such as where trailer angle is determined from images captured by camera **218**.

(77) Mast Mounted LIDAR

(78) In certain embodiments, LIDAR **220** may be mounted to an extensible mast on tractor **104**, so that it may be raised to an elevation sufficient for it capture trailer **106** and allow current trailer angle **233** to be determined. For example, the mast may be extended after tractor **104** couples with trailer **106** and while tractor **104** is stationary, to reduce loads on the extended mast. After LIDAR data is captured, the mast may be retracted before tractor **104** moves (e.g., to remain below vehicle max height limits).

(79) FIG. **16** is a front perspective view of tractor **104** coupled with trailer **106**, showing a LIDAR **220(3)** mounted to an extensible mast **1620**. In certain embodiments, mast **1620** incorporates upward facing sensors, a range finder, or similar sensors, to warn and prevent the extending mast from colliding with overhead obstructions. Further, mast **1620** may be stabilized by guy wires (not shown) that deploy from damped retracting reels and thereby reduce sway of LIDAR **220(3)** when mast **1620** is extended. Similarly, horizontal extending arms may be included to extend one or more sensors (e.g., camera **218** and/or LIDAR **220**) horizontally. By raising the sensors above the height of trailer **106**, or away from the sides of trailer **106**, the sensors are not blocked by trailer **106** and may thereby gain an improved perspective of a position of trailer **106** relative to tractor **104**.

(80) By way of example, mast **1620** may be implemented as a Rolatube™ (however, other extenders may also be used) that extends vertically, horizontally, or at some other angle, from tractor **104** to provide one or more sensors with a better (e.g., bird's eye) view of trailer **106**. For example, the Rolatube may provide an extension between ten and twenty feet that would allow a sensor (e.g., camera **218** and/or LIDAR **220**) a view around the side of trailer **106** at just about any backing position and angle. In certain embodiments, the extensive mast is mounted on a turret with continuous rotation about a vertical axis, and further, a second actuator may transition the turret from horizontal to vertical. The turret may include slip ring connections for power and signal.

(81) Additionally, a motorized element at the end of the Rolatube could be used with the turret to position one or more sensors at the edge or back of a trailer such that they may look back.

(82) Magnetically Coupled Trailer Angle Encoder

(83) FIG. **17** is a schematic diagram showing one example magnetically coupled trailer angle encoder **1700** positioned beneath FW **222** of tractor **104** prior to tractor **104** coupling (hitching) with trailer **106**. Trailer angle encoder **1700** includes a spring plate **1702**, a two-axis positioning table **1704**, a bearing **1706**, and optical encoder **1708**, a magnet **1710**, and a clearance and cleaning block **1712**. FIG. **18** is a schematic showing magnetically coupled trailer angle encoder **1700** of FIG. **17** magnetically coupled with kingpin **308** of trailer **106** that is hitched to tractor **104**. FIGS. **19**, **20** and **21** shows side, top and front elevations, respectively, of spring plate **1702** of FIGS. **17** and **18**. FIG. **22** is a top view of magnetically coupled trailer angle encoder **1700**. FIGS. **23**, **24**, and **25** are front, side, and top elevations, respectively, of two-axis positioning table **1710** of FIGS. **17** and **18**. FIGS. **17-25** are best viewed together with the following description.

(84) Trailer angle encoder **1700** is one example implementation of trailer angle encoder **204** of FIG. **2**. Trailer angle encoder **1700** measures trailer angle of trailer **106** using a magnetically coupled optical encoder with kingpin **308** and thereby capturing its rotational movements relative to tractor

104. Spring plate **1702** is made from spring steel for example and attaches, at a first end **1703**, to FW **222**. In certain embodiments, first end **1703** of spring plate **1702** includes structure and/or fasteners that attach to a flange or other structure beneath and towards the rear (relative to tractor **104**) of FW **222** without obstructing coupling of FW **222** with kingpin **308** of trailer **106**. In embodiments shown in FIGS. **19-21**, plate **1702** may be shaped to attach either side of a rear opening of FW **222** that receives kingpin **308**.

(85) Two-axis positioning table **1704** is attached to a second end **1705** of spring plate **1702**, opposite to first end **1703**, whereby spring plate **1702** positions two-axis positioning table **1704** beneath locking jaws **810** of FW **222**. Two-axis positioning table **1710** movably supports bearing **1706** and a shaft of optical encoder **1708** passes through bearing **1706** and rigidly attaches to magnet **1710**. Accordingly, magnet **1710** is positioned beneath locking jaws **810** such that it is attached to kingpin **308** when kingpin **308** is captured by locking jaws **810**.

(86) Clearance and cleaning block **1712** is a block with slopes forming a ridge and is positioned immediately behind two-axis positioning table **1704**. In certain embodiments, block **1712** is solid and made of durable plastic or rubber with a height **1713** that is greater than a height **1711** of a top surface of magnet **1710**, with respect to spring plate **1702**. Block **1712** has two functions: (a) it cleans a bottom surface of kingpin **308** as tractor **104** moves beneath trailer **106** during hitching, and (b) it depresses spring plate **1702** as kingpin **308** moved across it and forces it downward, thereby preventing kingpin **308** from impacting a side of magnet **1710**. As tractor **104** moves under trailer **106**, block **1712** wipes dirt from the bottom surface of kingpin **308** and moves off of kingpin **308** as kingpin **308** is seated within FW **222** and locked into place by locking jaws **810**. As kingpin **308** moves off block **1712**, spring plate **1702** returns to its nominal position and an upper surface of magnet **1710** contacts the (cleaned) lower surface of kingpin **308**.

(87) Since kingpin **308** is steel, magnet **1710** magnetically adheres to kingpin **308**. Two-axis positioning table **1710** allows minor offsets in alignment of the shaft of optical encoder **1708** and the rotational center of kingpin **308**, and for movement of kingpin **308** relative to tractor **104** during maneuvering of trailer **106**. For example, forces applied by tractor **104** to move trailer **106** may result in small lateral movements of kingpin **308** relative to FW **222**.

(88) A magnetic force attracts magnet **110** to kingpin **308** resulting in friction between the bottom surface of kingpin **308** and a top surface of magnet **1710**. Rotation of kingpin **308** relative to FW **222** (e.g., tractor **104**) results in an angular force applied to magnet **1710** that is greater than any combined angular resistance of bearing **1706** and optical encoder **1708**. Accordingly, rotation of kingpin **308** results in rotation of optical encoder **1708**.

(89) As shown in FIGS. **19-21**, first end **1703** of spring plate **1702** may form two folded lugs **2002** and **2004** that are spaced to attach to the structure beneath and to the rear of FW **222** without blocking ingress of kingpin **308** into locking jaws **810**. Spring plate **1702** may also form a notch **2006** at second end **1705** that receives optical encoder **1708**, which extends below two-axis positioning table **1710** and allows electrical connection therewith. Notch **2006** may have other shapes and sizes without departing from the scope hereof.

(90) As shown in FIGS. **22-25**, two-axis positioning table **1710** has a top platform **2202** that forms a first aperture **2502** and four mounting holes **2504(1)-(4)** for coupling with optical encoder **1708**. Top platform **2202** also has four mounting holes **2506(1)-(4)** for attaching bearing **1706** to top platform **2202**. Top platform **2202** slidably attaches to two parallel transverse rods **2302** that allow top platform **2202** to slide as indicated by arrow **2402**. A middle structure **2304** supports transverse rods **2302** and attaches to two parallel transverse rods **2306** that are perpendicular to transverse rods **2302** and that allows middle structure **2304** to move as indicated by arrow **2310**. Transverse rods **2306** are supported by a bottom structure **2308** that fixedly attaches to spring plate **1702**. Accordingly, relative to spring plate **1702**, top platform **2202** has lateral movement, indicated by arrows **2402** and **2310**, but is unable to rotate.

(91) In certain embodiments, as shown in FIG. **23**, two-axis positioning table **1710** includes two

springs **2312** and **2314** that each have first ends attached near a center of one side of middle structure **2304** and opposite ends attaches to opposite ends, respectively, of the same side of bottom structure **2308**. Springs **2312** and **2314** operates to center middle structure **2304** on bottom structure **2308** when no forces are applied to top platform **2202**. Forces applied by springs **2312** and **2314** are less than a force required to move magnet **1710** against kingpin **308**. That is, springs **2312** and **2314** center middle structure **2304** on bottom structure **2308** until magnet couples with kingpin **308**.

(92) Changes may be made in the above methods and systems without departing from the scope hereof. It should thus be noted that the matter contained in the above description or shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. The following claims are intended to cover all generic and specific features described herein, as well as all statements of the scope of the present method and system, which, as a matter of language, might be said to fall therebetween.

(93) Combination of Features

(94) Features described above as well as those claimed below may be combined in various ways without departing from the scope hereof. The following enumerated examples illustrate some possible, non-limiting combinations: (A1) A trailer angle encoder for determining an angle between a tractor and a trailer coupled thereto includes: an arm coupled at a pivot with a flange located beneath a fifth-wheel of the tractor; an optical encoder positioned at a first end of the arm and having a rotatable shaft with a mechanical coupler; and wherein the arm is positioned to mechanically couple the mechanical coupler with a kingpin of the trailer. (A2) The embodiment (A1) further including a solenoid mechanically coupled with the flange and operable to move a second end of the arm against a spring. (A3) In either one of embodiments (A1) or (A2), the solenoid causing the arm to disengage the mechanical coupler from the kingpin, and the spring causing the arm to engage the mechanical coupler with the kingpin when the solenoid is deactivated. (A4) In any one of embodiments (A1)-(A3), the solenoid causing the arm to engage the mechanical coupler with the kingpin. (A5) In any one of embodiments (A1)-(A4), the mechanical coupler including: a disc mechanically coupled to the rotatable shaft; and a plurality of pins distributed around a first side of the disc and each pin having a point that mechanically engages with a bottom surface of the kingpin. (A6) In any one of embodiments (A1)-(A5), the mechanical coupler including a conical adapter fixedly coupled with the rotatable shaft and having a conical shape that tapers internally from a first diameter, nearest the optical encoder, smaller than a diameter of a flange of the kingpin, to a second diameter greater than the diameter of the flange. (A7) In any one of embodiments (A1)-(A6), the conical adapter being at least partially formed of a flexible material that mechanically grips the flange when pressed there against. (A8) In any one of embodiments (A1)-(A7), the arm including: a first plate forming the first end; a second plate coupled with the pivot; at least one longitudinal shaft coupling the first plate with the second plate and allowing longitudinal movement of the first plate relative to the second plate; and the longitudinal movement allows the conical adapter to longitudinally center on the flange. (A9) Any one of embodiments (A1)-(A8) further including at least two springs positioned on the longitudinal shaft to return the first plate to a nominal longitudinally central position when the conical adapter disengages with the flange. (A10) In any one of embodiments (A1)-(A9), the pivot including a lateral shaft that further allows lateral movement of the second plate, where in the lateral movement allows the conical adapter to laterally center on the flange. (A11) Any one of embodiments (A1)-(A10) further including at least two springs positioned on the lateral shaft to return the second plate to a nominal laterally central position when the conical adapter disengages with the flange. (A12) In any one of embodiments (A1)-(A11), the first plate forming an aperture for securing the optical encoder. (B1) A method for determining an angle between a tractor and a trailer that are coupled together includes: controlling, from a controller of the tractor, the tractor to pull the trailer a short distance; determining, from an optical encoder mounted on the tractor and mechanically coupled

with the trailer, a change in angle between the tractor and the trailer; and calculating the angle between the tractor and the trailer based on the change in angle. (B2) In embodiments of (B1), the step of calculating including evaluating the formula:

(95) $\theta = \phi + \theta_0 - \Delta S + \sin^{-1}(-\frac{L_{sub.2}}{S})$ where ϕ is a heading of the tractor, θ is a heading of the trailer, and $L_{sub.2}$ is a length of the trailer, and ΔS is a distance moved. (C1) A trailer angle encoder for determining an angle between a tractor and a trailer coupled thereto includes: a spring plate for coupling at a first end with an underside of a fifth-wheel of the tractor; an optical encoder attached to the spring plate; a magnet mounted to a rotatable shaft of the optical encoder; and a clearance and cleaning block positioned on the spring plate to interact with a bottom surface of a kingpin of the trailer during hitching of the tractor to the trailer; wherein the magnet magnetically couples with the bottom surface of the kingpin when the tractor is hitched to the trailer. (C2) In embodiments of (C1), the spring plate positioning the magnet beneath the kingpin. (C3) Either one of embodiments (C1) or (C2) further including a two-axis positioning table mounted to the spring plate and having a platform that supports the optical encoder. (C4) In any one of embodiments (C1)-(C3), the platform providing limited two-axis movement of the optical encoder and the magnet relative to the fifth-wheel. (C5) Any one of embodiments (C1)-(C4) further including a bearing supported by the platform and coupled with the rotatable shaft of the optical encoder. (C6) Any one of embodiments (C1)-(C5) further including at least two springs to cause the platform to return to a nominal laterally central position when the magnet is not magnetically coupled with the kingpin. (C7) In any one of embodiments (C1)-(C6), the clearance and cleaning block having a height greater than a height of a top surface of the magnet and causing the kingpin to depress the clearance and cleaning block during hitching of the tractor to the trailer such that the spring plate is flexed downwards to prevent the kingpin contacting the side of the magnet. (C8) In any one of embodiments (C1)-(C7), the clearance and cleaning block being positioned to stop interaction of the magnet with the kingpin prior to capture of the kingpin by the fifth-wheel. (C9) In any one of embodiments (C1)-(C8), the interaction of the clearance and cleaning block with the bottom surface of the kingpin wiping dirt from the bottom of the kingpin. (D1) A software product includes instructions, stored on non-transitory computer-readable media, wherein the instructions, when executed by a processor, perform steps for determining an angle between a tractor and a trailer that are coupled together, includes: instructions for controlling, from a controller of the tractor, the tractor to pull the trailer a short distance; instructions for determining, from an optical encoder mounted on the tractor and mechanically coupled with the trailer, a change in angle between the tractor and the trailer; and instructions for calculating the angle between the tractor and the trailer based on the change in angle. (E1) A method for determining an angle between a tractor and a trailer that are coupled together includes: capturing, within a controller of the tractor, a point cloud using a rear facing LIDAR positioned on the tractor; converting points of the point cloud corresponding to front corners of the trailer to coordinate form; and calculating the angle between the tractor and the trailer based upon the coordinates of the front corners of the trailer. (E2) In embodiments of (E1), the step of converting including converting the points to coordinate form such that:

(96) $Y = M \cdot x + b$, $Y = [x[1]] \begin{bmatrix} M \\ b \end{bmatrix}$, and $\begin{bmatrix} M \\ b \end{bmatrix} = ([x[1]]^T [x[1]])^{-1} [x[1]]^T Y$ where x , Y is a

coordinate, and M and b are scalars. (F1) A software product includes instructions, stored on non-transitory computer-readable media, wherein the instructions, when executed by a processor, perform steps for determining an angle between a tractor and a trailer that are coupled together, includes: instructions for capturing, within a controller of the tractor, a point cloud using a rear facing LIDAR positioned on the tractor; instructions for converting points of the point cloud corresponding to front corners of the trailer to coordinate form; and instructions for calculating the

angle between the tractor and the trailer based upon the coordinates of the front corners of the trailer.

Claims

1. A method for determining a LIDAR-based angle between an autonomous tractor and a trailer that are coupled together, comprising: capturing, within a controller of the tractor, a point cloud using a rear facing LIDAR positioned on the tractor; converting points of the point cloud corresponding to front corners of the trailer to coordinate form; and calculating the LIDAR-based angle between the tractor and the trailer based upon the coordinates of the front corners of the trailer.
2. The method of claim 1, the converting comprising converting the points to coordinate form such that: $Y = M \cdot x + b$, $Y = [x[1]] \begin{bmatrix} M \\ b \end{bmatrix}$, and $\begin{bmatrix} M \\ b \end{bmatrix} = ([x[1]]^T [x[1]])^{-1} [x[1]]^T Y$, where x , Y is a coordinate, and M and b are scalars.
3. The method of claim 1, further comprising determining an encoder-based angle between the autonomous tractor and the trailer by: controlling, from the controller of the tractor, the tractor to pull the trailer a short distance; determining, from an encoder mounted on the tractor and mechanically coupled with the trailer, a change in angle between the tractor and the trailer; and calculating the encoder-based angle between the tractor and the trailer based on the change in angle.
4. The method of claim 3, the calculating the encoder-based angle comprising evaluating a formula: $\phi = \phi_0 + \theta_0 - \sin^{-1}(-\frac{L_2}{\Delta S})$, where ϕ is a heading of the tractor, θ is a heading of the trailer, and L_2 is a length of the trailer, and ΔS is a distance moved.
5. The method of claim 3, further comprising calculating a current trailer angle using both the encoder-based angle and the LIDAR-based angle.
6. The method of claim 5, the calculating the current trailer angle using both the encoder-based angle and the LIDAR-based angle including implementing a Kalman filter to process the LIDAR-based angle and the change in angle to generate the current trailer angle.
7. The method of claim 5, the calculating the current trailer angle using both the encoder-based angle and the LIDAR-based angle including implementing a Kalman filter that estimates initial bias in the encoder-based angle based on the LIDAR-based angle.
8. The method of claim 5, the calculating a current trailer angle further using image data captured by a camera.
9. The method of claim 1, further comprising calculating a current trailer angle using the LIDAR-based angle and images captured by a camera.
10. A system determining a LIDAR-based angle between an autonomous tractor and a trailer that are coupled together, comprising: a LIDAR for capturing a point cloud about the tractor, the LIDAR being rear-facing with respect to the tractor and the point cloud including points corresponding to front corners of the trailer; and, a controller configured to: convert the points of the point cloud corresponding to front corners of the trailer to coordinate form, and calculate the LIDAR-based angle between the tractor and the trailer based upon the coordinates of the front corners of the trailer.
11. The system of claim 10, the LIDAR being mounted on a cap portion of the tractor.
12. The system of claim 10, the LIDAR being mounted on an extensible mast.
13. The system of claim 12, further comprising an upward-facing sensor that provide mast-data used by the controller to prevent extending the extensible mast into overhead obstructions identified within the mast-data.
14. The system of claim 10, the controller further using one or both of image data and encoder data in conjunction with the coordinates to calculate a current trailer angle.

15. The system of claim 10, wherein the controller is configured to convert the points to coordinate form such that: $\theta_0 = \theta_0 + \theta_0 - \theta_0 + \sin^{-1}(-\frac{L_2}{S})$, where x, Y is a coordinate, and M and b are scalars.

16. The system of claim 10, the controller is further configured to calculate a current trailer angle using both encoder-based angle and the LIDAR-based angle.

17. The system of claim 16, the controller configured to calculate the current trailer angle using both the encoder-based angle and the LIDAR-based angle using at least a Kalman filter to process the LIDAR-based angle and a change in angle to generate the current trailer angle.

18. The system of claim 16, the controller configured to calculate the current trailer angle using a Kalman filter that estimates initial bias in the encoder-based angle based on the LIDAR-based angle.

19. The system of claim 10, the controller configured to calculate a current trailer angle further using image data captured by a camera.

20. The system of claim 19, the camera being mounted on the extendable mast.
